

Influence of smoking and plasma factors on patency of femoropopliteal vein grafts.

SCIFACT-EVIDENCE-12991445

OBJECTIVE To determine the effects of smoking, plasma lipids, lipoproteins, apolipoproteins, and fibrinogen on the patency of saphenous vein femoropopliteal bypass grafts at one year. **DESIGN** Prospective study of patients with saphenous vein femoropopliteal bypass grafts entered into a multicentre trial. **SETTING** Surgical wards, outpatient clinics, and home visits coordinated by two tertiary referral centres in London and Birmingham. **PATIENTS** 157 Patients (mean age 66.6 (SD 8.2) years), 113 with patent grafts and 44 with occluded grafts one year after bypass. **MAIN OUTCOME MEASURE** Cumulative percentage patency at one year. **RESULTS** Markers for smoking (blood carboxyhaemoglobin concentration (p less than 0.05) and plasma thiocyanate concentration (p less than 0.01) and plasma concentrations of fibrinogen (p less than 0.001) and apolipoproteins AI (p less than 0.04) and (a) (p less than 0.05) were significantly higher in patients with occluded grafts. Serum cholesterol concentrations were significantly higher in patients with grafts that remained patent one year after bypass (p less than 0.005). Analysis of the smoking markers indicated that a quarter of patients (40) were untruthful in their claims to have stopped smoking. Based on smoking markers, patency of grafts in smokers was significantly lower at one year by life table analysis than in non-smokers (63% v 84%, p less than 0.02). Patency was significantly higher by life table analysis in patients with a plasma fibrinogen concentration below the median than in those with a concentration above (90% v 57%, p less than 0.0002). Surprisingly, increased plasma low density lipoprotein cholesterol concentration was significantly associated with improved patency at one year (85%) at values above the median compared with patency (only 68%) at values in the lower half of the range (p less than 0.02). **CONCLUSIONS** Plasma fibrinogen concentration was the most important variable predicting graft occlusion, followed by smoking markers. A more forceful approach is needed to stop patients smoking; therapeutic measures to improve patency of vein grafts should focus on decreasing plasma fibrinogen concentration rather than serum cholesterol concentration.